

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (ECE) (Credit System) Degree Examinations
 January 2023

19ECE51 – ARTIFICIAL INTELLIGENCE

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain production systems used for search algorithms.	6	L*2	1	1
	b) With a flow diagram, develop a solution for the missionaries and cannibals problem.	8	L2	1	1
	c) Contrast between informed and brute force search algorithms.	6	L2	2	1
2.	a) Explain the variants of hill climbing algorithms in detail.	8	L2	1	1
	b) With an example state in the tic-tac-toe game. Draw the tree diagram for four steps listing all options.	8	L3	1	1
	c) List any eight applications of AI in healthcare.	4	L2	2	1
3.	a) With a flow chart, explain the A* algorithm in detail.	8	L2	1	1
	b) List the regions of the objective function with a neat diagram.	4	L2	1	1
	c) List the steps in depth first search and breadth first search algorithms.	8	L2	2	1
Unit – II					
4.	a) Explain inheritable knowledge representation with a suitable example.	7	L2	3	1
	b) List and explain the steps in Natural Language Processing.	5	L2	4	1
	c) List the primitive acts of conceptual dependance theory.	8	L2	3	1
5.	a) With a neat diagram explain the knowledge representation framework.	8	L2	3	1
	b) Explain the general learning model used in NLP with a neat block diagram.	7	L2	4	1
	c) Explain augmented transition network with an example.	5	L2	3	1
6.	a) Explain how frame sets can be used as instances.	7	L2	4	1
	b) Explain about universal quantifiers and existential quantifiers.	5	L2	4	1
	c) Explain about logical reasoning and symbolic reasoning.	8	L2	3	1
Unit – III					
7.	a) List the steps in Minimax search procedure.	10	L2	4	1
	b) Explain the following terms in neural networks i) Feed forward neural network ii) multi-layer neural network iii) Back propagation algorithms	10	L2	5	1
8.	a) Explain any two algorithms used for non-linear planning.	10	L2	4	1
	b) List the components in the planning system.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
